

Linear Separability of Boolean Functions: Perceptron Results

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A single McCulloch–Pitts neuron ($o = \text{sgn}(\mathbf{w} \cdot \mathbf{x} - \theta)$) was trained with the learning rule ($\Delta \mathbf{w} = \eta(t - o)\mathbf{x}$, $\Delta \theta = -\eta(t - o)$) to test the fraction of linearly separable boolean functions. For $n = 2, 3$ all 2^{2^n} possible target functions were tested. For $n = 4, 5$ 10^4 target functions were each drawn randomly. A function was considered linearly separable if the weights classified every target value correct for the boolean function they were trained on inside the number of epochs.

n	Theoretical fraction	Computed fraction (mean)	Standard deviation
2	0.875000	0.875000	0.000000
3	0.406250	0.405273	0.001735
4	0.028717	0.027385	0.001528
5	0.000022	0.000005	0.000022

Table 1: Fraction of linearly separable Boolean functions of n variables. Theoretical fraction is the known count of linearly separable functions (14, 104, 1882, 94572 for $n = 2, 3, 4, 5$) divided by 2^{2^n} . Computed values are the mean and sample standard deviation (ddof = 1) over $T = 20$ independent trials. See Table 2 for the parameters used.

Parameter	Value
Learning rate η	0.05
Weight initialisation	$w_i \sim \mathcal{N}(0, 1/n)$, redrawn per function
Threshold initialisation	$\theta = 0$, reset per function
Activation function	$\text{sgn}(b_m)$, if $b_m = 0$ then $\text{sgn}(b_m) = 1$
Max. training epochs per function	20
Update scheme	Asynchronous: one weight update per input pattern per epoch
Independent trials T	20
Functions tested, $n = 2, 3$	All 2^{2^n} (16 and 256)
Functions tested, $n = 4, 5$	10^4 random target functions per trial (with replacement)

Table 2: Parameters used to get the results in Table 1.

Discussion. For $n = 2$ the computed and theoretical fractions match exactly with zero variance: with only 4 input patterns, 20 epochs are always enough to converge. For $n = 3$ the computed value is slightly below theory (0.405273 vs. 0.406250, $\approx 0.56\sigma$); this small downward bias comes from the finite epoch budget – a few separable functions have a small margin and occasionally fail to fully converge within 20 epochs, so they get miscounted as non-separable (this can only push the estimate down, never up, since a truly non-separable function can never pass the check). For $n = 4, 5$ an additional source of uncertainty appears: only 10^4 of the 2^{16} or 2^{32} possible functions are sampled per trial, adding genuine Monte Carlo noise on top of the epoch-budget effect. Both results fall within $\approx 1\sigma$ of theory (0.87σ and 0.77σ). The $n = 5$ case is a rare-event estimate: the true fraction is only $\approx 2.2 \times 10^{-5}$, so across the whole experiment ($10^4 \times 20$ draws) only about 4–5 separable functions are expected in total, making relative fluctuations large even though the absolute deviation is small and unremarkable. **Conclusion:** all four measured fractions are consistent with the known theoretical values once the finite training budget and finite sample size are accounted for; the observed deviations are explained by these two effects rather than by an error in the learning rule or separability check.